

DR-2 sum-level-circle reduction – an unconditional additive-energy

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1. Purpose and scope

The decoupling route (dr2-cross-scale-induction v1.1) reduces unrestricted DR-2 to two cited ingredients of Bourgain–Demeter machinery, of which R2 (the multi-scale constant control) is the hard core of decoupling itself and is not reproducible here. This note steps off that route and proves, UNCONDITIONALLY and elementarily, the part that IS provable: an exact reduction of the sphere additive energy to a single geometric quantity, the occupancy of sum-level circles. It yields elementary DR-2 for a natural class and pins the precise obstruction for the rest.

2. The sum-level circle

Let $Q \subset S^2$ be finite, $|Q| = N$, and for $m \in \mathbb{R}^3$ let $r(m) = \#\{(a, b) \in Q^2 : a + b = m\}$, so that the additive energy is $E_+(Q) = \sum_m r(m)^2$ and $\sum_m r(m) = N^2$.

Geometric fact. If $a, b \in S^2$ and $a + b = m$ then

$$a \cdot m = a \cdot (a + b) = |a|^2 + a \cdot b = 1 + a \cdot b, \quad |m|^2 = |a|^2 + 2a \cdot b + |b|^2 = 2 + 2a \cdot b,$$

so $a \cdot m = |m|^2/2$, and likewise $b \cdot m = |m|^2/2$. Hence both summands lie on the *sum-level circle*

$$C_m := S^2 \cap \{x \in \mathbb{R}^3 : x \cdot m = \tfrac{1}{2}|m|^2\}.$$

For $m \neq 0$ this is a proper circle (a plane at distance $|m|/2 > 0$ from the origin meeting S^2 , nonempty iff $|m| \leq 2$); for $m = 0$ the defining plane degenerates to all of $\mathbb{R}^3$ and $C_0 = S^2$.

3. Lemma A (unconditional)

Lemma A. For every finite $Q \subset S^2$,

$$\boxed{E_+(Q) \leq (1 + T'(Q)) N^2}, \quad T'(Q) := \max_{\substack{m \neq 0 \\ r(m) > 0}} \#(Q \cap C_m).$$

Proof. Split $E_+(Q) = r(0)^2 + \sum_{m \neq 0} r(m)^2$.

(i) *The $m = 0$ term.* $r(0) = \#\{(a, b) \in Q^2 : a + b = 0\} = \#\{a \in Q : -a \in Q\} \leq N$, so $r(0)^2 \leq N^2$.

(ii) *The proper terms.* Fix $m \neq 0$ with $r(m) > 0$. Every pair (a, b) with $a + b = m$ has $a \in Q \cap C_m$ and $b = m - a$ determined by a , so $r(m) = \#\{a \in Q \cap C_m : m - a \in Q\} \leq \#(Q \cap C_m) \leq T'(Q)$. Therefore

$$\sum_{m \neq 0} r(m)^2 \leq T'(Q) \sum_{m \neq 0} r(m) \leq T'(Q) N^2.$$

Adding (i) and (ii) gives $E_+(Q) \leq N^2 + T'(Q)N^2 = (1 + T'(Q))N^2$. $\square$

The split is sharp and necessary: for N evenly spaced points on a great circle the antipodal term contributes $r(0)^2 = N^2$ and the proper circles have occupancy $T' = 2$, so the bound reads $(1 + 2)N^2 = 3N^2$ against the exact $E_+ = 2.95 N^2$ (dr2_circle_richness.py: N^2 antipodal + $2N^2$ proper).

4. Consequences

Corollary 1 (elementary unconditional DR-2 for bounded richness). If $T'(Q) = O(1)$ then $E_+(Q) = O(N^2)$ – DR-2 holds with NO decoupling and NO conjecture. The script confirms $T' = 2$ for both random sphere sets ($E_+ = 1.98 N^2$) and the great circle ($E_+ = 2.95 N^2$); these natural classes satisfy DR-2 by Lemma A alone.

Corollary 2 (reduction of general DR-2). For arbitrary finite Q ,

$$T'(Q) \lesssim_\epsilon N^\epsilon \implies E_+(Q) \lesssim_\epsilon N^{2+\epsilon} \quad (\text{DR-2}).$$

Thus DR-2 follows from a bound on a single elementary quantity: the maximum occupancy of a sum-level circle. This is a concrete realisation of the DR2-SHARE “carrier richness” $\chi(P)$ with an EXACT constant $(1 + T')$, sharper than the polylog framing.

Scope honesty (the condition is sufficient, not tight). Corollary 2 is one-directional. The engineered latitude circle $z = z_0$ has a single rich sum-circle ($m^* = (0, 0, 2z_0)$, all N points on it, $r(m^*) = N$), so $T' = N$, yet $E_+ \approx 3N^2$ still satisfies DR-2. Hence bounding T' is potentially STRONGER than DR-2, and Lemma A does not by itself make the general problem easier; it isolates rich sum-circles as the only mechanism by which E_+ can exceed $O(N^2)$ and supplies the unconditional bound for the rest.

5. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3, code-discipline 6)

Four concrete objections, found before submission:

- (α) *“Does this close DR-2 or advance its tier?”* NO, and the note must not be read that way. Lemma A and Corollary 1 are T7 UNCONDITIONAL but cover only the bounded-richness class; the unrestricted DR-2 tier is UNCHANGED (T4+ via the decoupling route). No gate/tier/hypothesis flip.
- (β) *“The $m = 0$ circle is all of S^2 , so T' would be N always, making the bound trivial N^3 .”* DISMISSED: $m = 0$ is excluded from the max defining T' and handled separately in step (i), where $r(0) \leq N$ gives the additive “+1”. The great-circle check (tight at $3N^2$) confirms the split is correct, not a loophole.
- (γ) *“ T' can be N while E_+ is $O(N^2)$ (rich latitude), so the reduction is vacuous.”* VALID and RECORDED as the scope-honesty paragraph: Corollary 2 is sufficient, not tight. Its value is (a) the unconditional bound for $T' = O(1)$ sets and (b) pinning rich sum-circles as the sole obstruction – not a claim that $T' \lesssim N^\epsilon$ is equivalent to DR-2.
- (δ) *“Is this just the known carrier-richness reduction re-labelled?”* PARTIALLY: it makes the carrier-richness concrete (the precise quantity T' , exact constant $1 + T'$, via the sum-circle/involution identity) and proves the easy direction rigorously and unconditionally,

whereas the prior PSM reduction (dr2-pencil-rigidity-reduction v1.0) was a T3 sketch. Whether $T' \lesssim N^\epsilon$ is equivalent to the PSM conjecture is NOT asserted here.

Quantitative sanity check (mandatory). Estimator exactness: random $E_+ = 2N^2 - N$ verified for $N = 16, 32, 64$. Lemma assert: $E_+ \leq (1 + T')N^2$ holds for all four configs (the proof check; a violation would refute Lemma A). Limit/tightness: great circle saturates at $3N^2 = (1+2)N^2$ (the $m = 0$ split is necessary). Magnitude/direction: $T' = 2$ (random, great circle) $\rightarrow 8$ (cap grid) $\rightarrow N$ (rich circle), all ≥ 1 , upper bound, all terms positive. Reproducibility: T' of random sets stays 2 for $N = 32, 64, 128$.

6. Result footer

Result ID:	R-025 / DR-2 sum-level-circle reduction
Precise statement:	For finite Q in S^2 with $N= Q $, $E_+(Q) \leq (1+T'(Q))N^2$, where $T'(Q) = \max$ occupancy of a proper ($m \neq 0$) sum-level circle $C_m = S^2 \cap \{x.m = m ^2/2\}$. Corollary 1: $T'=0(1) \Rightarrow E_+=0(N^2)$ (DR-2, unconditional). Corollary 2: $T' \leq \epsilon \Rightarrow E_+ \leq N^{2+\epsilon}$.
Scope:	Unrestricted-class sphere additive energy; the DR2-SHARE carrier-richness obstruction, made exact.
Dependencies:	NONE (unconditional; elementary geometry + Cauchy-Schwarz via max). Decoupling-free and conjecture-free.
Evidence grade:	Lemma A + Corollary 1: T7 PROVED unconditional (proof self-tested by the lemma assert). Corollary 2: T7 reduction (valid sufficient condition). General DR-2: unchanged OPEN.
Reproduction command:	python codes/vacuum/dr2_circle_richness.py
Expected output:	5/5 PASS; $E_+ \leq (1+T')N^2$ for every config; $T' = 2$ (random/great-circle) up to N (rich latitude circle).
Falsification gate:	A finite Q in S^2 with $E_+(Q) > (1+T'(Q))N^2$ would refute Lemma A; not observed (and impossible by proof).
Tier before / after:	NEW unconditional T7 sub-result; unrestricted DR-2 tier UNCHANGED (T4+). NO flip. B5 T5, B1 T6, H-ADM-COH retained, DR2-SHARE OPEN.
No-overclaim statement:	NOT a DR-2 closure and NOT a tier advance for the general problem. Lemma A is unconditional but the reduction (Cor 2) is sufficient, not tight; bounding T' for arbitrary Q remains the open carrier-richness problem. No gate/tier/hypothesis action.
Next required action:	Either bound $T'(Q)$ for arbitrary Q (open; = carrier richness), or accept the decoupling route's R1+R2 by operator sign-off. Operator authorises any flip.